9. Rule-Based POS Tagging Using Regular Expressions

Aim: To implement a rule-based POS tagging system using regular expressions in Python.

Procedure:

1. Import the re module.
2. Define a list of words.
3. Create regular expression rules.
4. Match each word against the rules.
5. Display the assigned POS tags.

Code:

```
import re

words = ["playing", "quickly", "beautiful", "cat"]

for word in words:
    if re.search(r"ly$", word):
        tag = "RB"
    elif re.search(r"ing$", word):
        tag = "VBG"
    elif re.search(r"ful$", word):
        tag = "JJ"
    else:
        tag = "NN"
    print(word, ":", tag)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/9.py"
playing : VBG
quickly : RB
beautiful : JJ
cat : NN
```

Result: A rule-based POS tagging system was successfully implemented using regular expressions.